

KARTHIKEYAN P

Software Developer | AI/ML Engineer

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OBJECTIVE

Results-driven Software Developer with hands-on experience in Python, Java, Spring Boot, AI/ML development, and database-driven applications. Skilled in building intelligent automation systems, multi-agent AI pipelines, and full-stack web applications. Seeking a full-time software engineering role to deliver scalable, efficient, and innovative solutions.

EDUCATION

B.E. – Computer Science Engineering <i>Government College of Engineering, Erode</i>	2023 – Present CGPA: 8.14 / 10.00
Diploma – Computer Science Engineering <i>Muthayammal Polytechnic College, Namakkal</i>	2020 – 2023 96%

INTERNSHIP

End-User Tester Intern <i>The Context Lab – Multi-Agent Orchestration & AI Evaluation Platform</i>	Mar 2026
- Tested and validated agent workflows on a multi-agent orchestration and AI evaluation platform. - Identified, documented, and reported bugs to the developer team, ensuring accurate and reliable evaluation results. - Gained practical experience in AI agent testing, prompt evaluation, and quality assurance workflows.	

PROJECTS

KAI – AI Voice Assistant Agent <i>AI/ML Project Python, Ollama, Whisper STT, LLM, ChromaDB</i>	2026 GCE, Erode
- Architected a Jarvis-style personal AI assistant with a 4-layer modular architecture: Voice Engine, AI Brain, Tool Layer, and Memory Layer. - Implemented multi-agent LLM pipeline using local models (Phi-3, LLaMA) via Ollama for zero cloud dependency inference. - Built voice-controlled OS automation layer with system controls for Wi-Fi, volume, brightness, screenshots, and app launching. - Integrated RAG-based persistent memory using ChromaDB for long-term knowledge retention and context-aware responses.	
BroCodex – AI Code Correction & Optimization Pipeline <i>AI Project Python, Multi-Agent AI, LLM</i>	2026 GCE, Erode
- Built a multi-agent-to-single-agent pipeline that autonomously detects, corrects, and optimizes code across an entire codebase. - Designed a selector agent that evaluates and merges outputs from multiple specialized AI agents into one production-ready codebase.	

- Automated project dependency installation, enabling the system to self-configure any Python project environment without manual setup.

VendX – AI Price Analysis Tool

2026

Web Scraping / Data Analytics / AI | Python, BeautifulSoup, Pandas

GCE, Erode

- Built a Python-based web scraping pipeline to extract real-time product data, pricing, and stock availability across e-commerce platforms.
- Applied data analytics and AI models to identify best-selling price bands and generate business-facing product insight reports.
- Handled anti-scraping mechanisms using rotating headers and request throttling for reliable data extraction at scale.

Connect Application

2025

Full Stack Web Application | Java, Spring Boot, MySQL, REST API

GCE, Erode

- Developed a student-alumni networking platform enabling mentorship, referrals, and career guidance monitored by admin users.
- Implemented secure role-based authentication and authorization across student, alumni, and admin roles using Spring Security.
- Built RESTful APIs for profile management, connection requests, and real-time messaging between students and professionals.

TECHNICAL SKILLS

Programming Languages	Languages	Python, Java, JavaScript, HTML, CSS
AI / ML		LLM Integration, Multi-Agent Systems, NLP, RAG, Machine Learning, Neural Networks, Whisper STT
Frameworks & Tools		Spring Boot, Flask, React (Basic), Servlets
Databases		MySQL, PostgreSQL, ChromaDB
Libraries		Pandas, BeautifulSoup, Selenium, Ollama, LangChain
Deployment		AWS EC2, Vercel, Git, GitHub
Concepts		REST API, OOP, SDLC, Agile, Data Structures, Algorithms

CERTIFICATIONS

- Web Scraping with Python – IBM
- Full Stack Web Development using Amazon EC2 – IBM
- Full Stack Certification – Infrix

ACHIEVEMENTS

- Participated in multiple hackathons and paper presentations at national-level technical events.
- Secured 96% in Diploma in Computer Science Engineering – Academic Excellence.
- Built 5+ full-stack and AI/ML projects independently during undergraduate studies.

SOFT SKILLS

Problem Solving | Debugging | Analytical Thinking | Team Collaboration | Self-Learning